

What is a proposition?

a declarative statement that is either TRUE or FALSE.

Examples,

\* Sunday is a holiday ✓

\* 27 July 2026 is the first day of the semester ✓

\* Water is a solid ✓

\* What is the height of Everest? ✗

\*  $x^2 + y^2 = z^2$  ✗

Use of propositional variables

$P, Q$

Logical operators

(i) Not ( $\neg$ )

Truth Table (here  $p$  is a proposition)

| $p$ | $\neg p$ |
|-----|----------|
| T   | F        |
| F   | T        |

English Examples

$p$ : It is raining today

$\neg p$ : It is not raining today..

## ② Logical AND or conjunction ( $\wedge$ )

Two propositions  $P, Q$ .

$P \wedge Q$ : TRUE only when both  $P$  &  $Q$  are TRUE.

TRUTH TABLE

| $P$ | $Q$ | $P \wedge Q$ |
|-----|-----|--------------|
| F   | F   | F            |
| F   | T   | F            |
| T   | F   | F            |
| T   | T   | T            |

## Logical OR or Disjunction ( $\vee$ )

$P, Q$  are propositions

$P \vee Q$  is TRUE when  $P$  or  $Q$  is TRUE.

| $P$ | $Q$ | $P \vee Q$ |
|-----|-----|------------|
| F   | F   | F          |
| F   | T   | T          |
| T   | F   | T          |
| T   | T   | T          |

## Exclusive OR (XOR) ( $\oplus$ )

$P \oplus Q$ : TRUE when exactly one of  $P$  or  $Q$  is TRUE.

| P | Q | $P \oplus Q$ |
|---|---|--------------|
| F | F | F            |
| F | T | T            |
| T | F | T            |
| T | T | F            |

Conditional statements or implications

$P, Q$  are two propositions.

$P \rightarrow Q$  (" $P$  implies  $Q$ " or "if  $P$  then  $Q$ ")

The statement is FALSE when  $P$  is TRUE and  $Q$  is FALSE.

| $\neg P$ | $\neg Q$ | P | Q | $P \rightarrow Q$ | $\neg Q \rightarrow \neg P$ |
|----------|----------|---|---|-------------------|-----------------------------|
| T        | T        | F | F | T                 | T                           |
| T        | F        | F | T | T                 | T                           |
| F        | T        | T | F | F                 | F                           |
| F        | F        | T | T | T                 | T                           |

contrapositive of  $P \rightarrow Q$

$P$ : it rains today

$Q$ : we will not play cricket

$P \rightarrow Q$ : If it rains today,  
then we will not play  
cricket.

Other English constructs for  $P \rightarrow Q$

\* if  $P$  then  $Q$

\*  $P$  implies  $Q$

\*  $P$  only if  $Q$

\*  $Q$  whenever  $P$

\*  $Q$  unless  $\neg P$

\*  $P$  is sufficient for  $Q$

\*  $Q$  is necessary for  $P$

\*  $Q$  when  $P$

\*  $Q$  follows from  $P$

| P | Q | $\neg P$ | $Q$ unless $\neg P$ |
|---|---|----------|---------------------|
| F | F | T        | T                   |
| F | T | T        | T                   |
| T | F | F        | F                   |
| T | T | F        | T                   |

|   |   |   |   |
|---|---|---|---|
| T | T | F | T |
|---|---|---|---|

## Definitions of Converse, Contrapositive, and Inverse

Given a conditional statement

" $p \rightarrow q$ ", its

Converse:  $q \rightarrow p$

Contrapositive:  $\neg q \rightarrow \neg p$

Inverse:  $\neg p \rightarrow \neg q$

Truth table

Contrapositive of converse is inverse

| $p$ | $q$ | $\neg p$ | $\neg q$ | $p \rightarrow q$ | Converse<br>$q \rightarrow p$ | Contrapositive<br>$\neg q \rightarrow \neg p$ | Inverse<br>$\neg p \rightarrow \neg q$ |
|-----|-----|----------|----------|-------------------|-------------------------------|-----------------------------------------------|----------------------------------------|
| F   | F   | T        | T        | T                 | T                             | T                                             | T                                      |
| F   | T   | T        | F        | T                 | F                             | T                                             | F                                      |
| T   | F   | F        | T        | F                 | T                             | F                                             | T                                      |
| T   | T   | F        | F        | T                 | T                             | T                                             | T                                      |

English examples

Implication: "If it rains, then the home team wins"

Contrapositive: "If the home team does not win,  
then it did not rain"  
↳ is not raining

Converse: "If the home team wins then it rains"

Converse: "If the home team wins, then it rains"

Inverse: "If it does not rain, then the home team does not win"

### Biconditional statements

$p, q$  are propositions

$p \leftrightarrow q$  :  
(or  $p$  if and only if  $q$ )

This is TRUE when both  $p$  and  $q$  are TRUE or when both  $p$  and  $q$  are FALSE.

| $p$ | $q$ | $p \leftrightarrow q$ |
|-----|-----|-----------------------|
| F   | F   | T                     |
| F   | T   | F                     |
| T   | F   | F                     |
| T   | T   | T                     |

### Compound propositions

Logicomix

①  $(p \vee \neg q) \rightarrow (p \wedge q)$

| P | Q | $\neg Q$ | $P \vee \neg Q$ | $P \wedge Q$ | $(P \vee \neg Q) \rightarrow (P \wedge Q)$ |
|---|---|----------|-----------------|--------------|--------------------------------------------|
| F | F | T        | T               | F            | F                                          |
| F | T | F        | F               | F            | T                                          |
| T | F | T        | T               | F            | F                                          |
| T | T | F        | T               | T            | T                                          |

Precedence of logical ops

| op                | precedence |
|-------------------|------------|
| $\neg$            | 1          |
| $\wedge$          | 2          |
| $\vee$            | 3          |
| $\rightarrow$     | 4          |
| $\leftrightarrow$ | 5          |

Translating English sentences

- ① "You can access the internet from campus only if you are a CS major or you are not a first year student"

$$P \rightarrow (Q \vee R)$$

- ② "You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years"

$$(q \wedge \neg r) \rightarrow p$$

| p | q | r |
|---|---|---|
| F | F | F |
| F | F | T |
| F | T | F |
| F | T | T |
| T | F | F |
| T | F | T |
| T | T | F |
| T | T | T |

Q 38 (page 25)

| Guess    |                                                           |
|----------|-----------------------------------------------------------|
| ① K ✓    | $\neg K$                                                  |
| Then V ✓ | Then H                                                    |
| $\neg R$ | $\neg V$                                                  |
| $\neg A$ | R                                                         |
| $\neg H$ | Contradiction as it violates $H \rightarrow (A \wedge K)$ |

$$K \vee H$$

$$R \oplus V$$

$$A \rightarrow R$$

$$V \leftrightarrow K$$

$$H \rightarrow \underline{(A \wedge K)}$$

K: Kevin is chatting  
H: Heather is chatting

Q 42 (zebra puzzle)

| Nationality | Drink | House | Job | Pet |
|-------------|-------|-------|-----|-----|
| E           |       | R     |     |     |
| S           |       |       |     | D   |

|          |          |                |   |   |
|----------|----------|----------------|---|---|
| <u>s</u> |          |                |   | d |
| j        |          |                | p |   |
| <u>i</u> | <u>t</u> |                |   |   |
| 2        |          | first<br>house |   |   |

2  
1 1 1 1 1 1 1 1

Q 39

$$b \rightarrow c$$

$$\neg(c \wedge g)$$

$$g \vee h$$

$$h \rightarrow \neg c$$

Guess.

Suppose b. X

Then c

Th

g

Contradicts

$$\neg(c \wedge g).$$

b: butler is telling  
truth

c: cook is telling  
the truth.

s: gardener ...

h: handyman ...

...

We conclude  $\neg b$ .

Assume c. Then Th, s.

Contradicts  $\neg(c \wedge g)$ .

We conclude  $\neg b, \neg c$ . Assume g.

Bitwise OR (AND)

x: 000111010

y: 010110101

Result OR

010111111



Bitwise OR

0 1 0 1 1 1 1 1

Bitwise AND

0 0 0 1 1 0 0 0

Bitwise XOR

0 1 0 0 0 1 1 1

## Propositional Equivalences

Two compound propositions are said to be "logically equivalent" if they have the same truth values in all possible cases.

Definition: A compound proposition is called a **TAUTOLOGY** if it is always TRUE, no matter what truth values the propositions that occur in it take.

Example 1

$$p \vee (\neg p)$$

| $p$ | $\neg p$ | $p \vee \neg p$ |
|-----|----------|-----------------|
| F   | T        | T               |
| T   | F        | T               |

Example 2

$$(p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q)$$

Logical equivalence in terms of tautology

Two compound propositions  $p$  and  $q$  are logically equivalent if  $p \leftrightarrow q$  is a

## TAUTOLOGY,

Defn. A Compound proposition is called a CONTRADICTION if it is always FALSE, no matter what truth values the propositions present in it take.

Example 1

$$p \wedge \neg p$$

Example 2

$$(p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$$

Q. Show that  $\neg(p \vee q) \equiv \neg p \wedge \neg q$  → notation for logical equivalence

| p | q | $\neg p$ | $\neg q$ | $\neg(p \vee q)$ | $\neg p \wedge \neg q$ |
|---|---|----------|----------|------------------|------------------------|
|   |   |          |          |                  |                        |
|   |   |          |          |                  |                        |
|   |   |          |          |                  |                        |
|   |   |          |          |                  |                        |
|   |   |          |          |                  |                        |

De Morgan's Law

$$(a) \quad \neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$(b) \quad \neg(p \wedge q) \equiv \neg p \vee \neg q$$

Q. Show that  $p \rightarrow q \equiv \neg p \vee q$

Draw the truth table and check.

F F

$p \rightarrow T$   
 $q \rightarrow F$

Show that

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$\begin{aligned} (p \rightarrow r) \wedge (q \rightarrow r) &\equiv (\neg p \vee r) \wedge (\neg q \vee r) && [\text{since } a \rightarrow b \equiv \neg a \vee b] \\ &\equiv (r \vee \neg p) \wedge (r \vee \neg q) && [\text{Commutativity}] \\ &\equiv r \vee (\neg p \wedge \neg q) && [\text{Distributivity}] \\ &\equiv r \vee \neg(p \vee q) && [\text{De Morgan}] \\ &\equiv \neg(p \vee q) \vee r && [\text{Commutativity}] \\ &\equiv (p \vee q) \rightarrow r \end{aligned}$$

$$\text{Q} \quad (p \rightarrow r) \vee (q \rightarrow r) \equiv \underbrace{\begin{matrix} T & T & F \\ (p \wedge q) \rightarrow r \end{matrix}}_{\text{only true when RHS is F}}$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (\neg p \vee r) \vee (\neg q \vee r)$$

$$\equiv \neg p \vee \neg q \vee r \quad \left( \begin{array}{l} \text{by associativity} \\ \& \\ \text{commutativity} \end{array} \right)$$

$$\equiv \neg(p \wedge q) \vee r$$

$$\equiv (p \wedge q) \rightarrow r$$

$$\text{Q. } \neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

$$\begin{aligned} \neg(p \leftrightarrow q) &\equiv \neg((p \wedge q) \vee (\neg p \wedge \neg q)) \\ &\equiv \neg(p \wedge q) \wedge \neg(\neg p \wedge \neg q) \quad (\text{De Morgan}) \\ &\equiv (\neg p \vee \neg q) \wedge (p \vee q) \\ &\equiv (p \rightarrow \neg q) \wedge (\neg q \rightarrow p) \quad [a \rightarrow b \equiv \neg a \vee b] \\ &\equiv p \leftrightarrow \neg q \end{aligned}$$

Q 19. <sup>show</sup>  $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$  is a tautology.

$$\begin{aligned} (\neg q \wedge (p \rightarrow q)) \rightarrow \neg p &\equiv (\neg q \wedge (\neg p \vee q)) \rightarrow \neg p \\ &\equiv ((\neg q \wedge \neg p) \vee (\neg q \wedge q)) \rightarrow \neg p \quad [a \rightarrow b \equiv \neg a \vee b] \\ &\equiv ((\neg q \wedge \neg p) \vee F) \rightarrow \neg p \quad [\text{Distr. law}] \\ &\equiv (\neg q \wedge \neg p) \rightarrow \neg p \quad (\text{since } \neg q \wedge q \equiv F) \end{aligned}$$

This is a tautology because the only case the above stmt is FALSE is when  $(\neg q \wedge \neg p)$  is TRUE and  $\neg p$  is FALSE. This cannot happen.

$$\neg ( \neg \neg \neg \neg \neg p \rightarrow b \equiv )$$

THIS CARDS. 10

$$\begin{aligned}
 &\rightarrow \equiv \neg (\neg q \wedge \neg p) \vee \neg p \quad \left[ \begin{array}{l} a \rightarrow b \equiv \\ \neg a \vee b \end{array} \right] \\
 &\equiv q \vee \overbrace{p \vee \neg p}^{\text{TRUE}} \\
 &\equiv \text{TRUE} \\
 &\quad \quad \quad \underline{\underline{\quad}}
 \end{aligned}$$

Q 70.  
Let  $1 \leq i, j \leq 9$ .

$n_{ij} = T$ , if  $(ij)$ -th cell in the  
sudoku puzzle has a  
number  
 $= F$ , otherwise

$$\bigwedge_{\substack{1 \leq i \leq 9, \\ 1 \leq j \leq 9}} n_{ij}$$

Predicates & Quantifiers

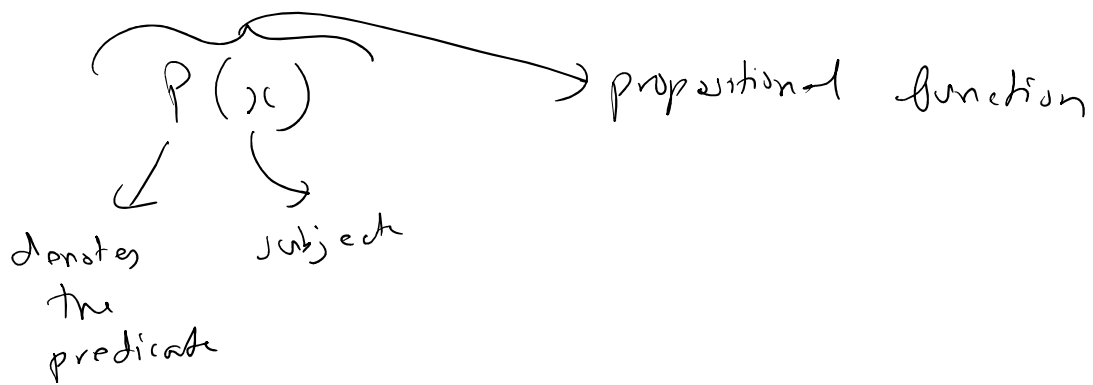
$$x > 3$$

↪ cannot be expressed in  
propositional logic

predicate logic is propositional logic with variables.

"  
 $x$  is greater than 5"  
subject      predicate

subject      predicate



Example Let  $P(x)$  denote " $x > 3$ ".  
 What are truth values of  $P(7)$ ,  $P(1)$ ,  $P(-8)$ ?  
T F F

Example Let  $P(x, y)$  denote " $x + y \geq 6$ ".

$P(3, 3)$  T

$P(7, -5)$  F

$P(5, 9)$  T